

Rotations characterised by $SU(2)$

Daniel Corper

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1 2d rotations

We have already established that moduli 1 complex numbers can be used to represent continuous rotations in 2 dimensions (see "A group theoretic approach to a matrix representation of the imaginary unit"). The natural next step is to ask whether there is any such representation for 3d rotations.

2 The symplectic group $Sp(1)$

The $Sp(2n, \mathbb{F})$ group is generally defined as the group of all $2n$ dimensional linear transformations over $\mathbb{F}$ that preserve a skew symmetric bilinear form J . More concisely:

$$Sp(2n, \mathbb{F}) = \{M_{2n \times 2n} : M^T J M = J\} \quad (1)$$

It happens that $Sp(1)$ forms the group of all unit *quaternions*. A quaternion is a generalisation to the complex number, usually written as:

$$a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} \quad (2)$$

Where $\mathbf{i} \neq \mathbf{j} \neq \mathbf{k}$ and $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$. This is similar to how $U(1)$ forms the group of all unit complex numbers. Just as unitary complex numbers can be used to represent a 2d rotation, quaternions can be used to represent a 3d rotation. However, it is not as simple as just saying that there exists an isomorphism between $Sp(1)$ and $SO(3)$.

3 A matrix representation of the unit quaternion

Whilst there does not exist an isomorphism between $Sp(1)$ and $SO(3)$, there does exist one between $Sp(1)$ and $SU(2)$, as some of you may have caught from (1). Following a similar process to the aforementioned document in section 1, we

find that the Lie algebra basis for $SU(2)$ under the fundamental representation is given by the Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3)$$

The matrices should form a direct representation of the purely imaginary quaternions, but they do not immediately square to the negative of the identity, to ensure this, we scale by $-i$:

$$\mathbf{i} = -i\sigma_x \quad \mathbf{j} = -i\sigma_y \quad \mathbf{k} = -i\sigma_z \quad (4)$$

4 Forming an isomorphism to $\mathfrak{so}(3)$

Some of you (especially physicists) may be familiar the isomorphism:

$$\mathfrak{su}(2) \cong \mathfrak{so}(3) \quad (5)$$

In that, the Lie algebra $\mathfrak{so}(3)$ is structured as:

$$[J_i, J_j] = \varepsilon_{ijk} J_k \quad (6)$$

The Lie algebra for $\mathfrak{su}(2)$ is very similar:

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk} \sigma_k \quad (7)$$

Therefore, whilst there is no isomorphism between the groups themselves, we can use these generators to specify a rotation in terms of $SU(2)$. Following the definition of a Lie algebra, a rotation is given by:¹

$$U = \exp(\theta J \cdot \hat{n}) \quad (8)$$

Where $\hat{n}$ is the unit vector specifying by which axis the rotation occurs. Because of the aforementioned Lie algebra isomorphism, that same rotation can be generated by scaled Pauli matrices, namely:

$$U = \exp\left(\frac{1}{2i}\sigma \cdot \hat{n}\right) \quad (9)$$

But when we actually carry out this matrix exponential, we obtain:

$$U = \cos(\theta/2)\mathbb{I}_2 + i\sin(\theta/2)\sigma \cdot \hat{n} \quad (10)$$

The rotation is half of what you'd expect for the same $SO(3)$ element. This follows intuitively because of how $SU(2)$ elements act on vectors, specifically:

$$V \rightarrow V' = U^\dagger V U \quad (11)$$

¹Another, perhaps more intuitive way of thinking about this, is that we can consider a 2d rotation as the exponentiation of i , we can also consider a 3d rotation as an exponentiation of a linear combination of the quaternion basis

So each element of $SU(2)$ contributes half of the rotation. You may also have noticed that there exists two elements of $SU(2)$ then for each element of $SO(3)$. Suppose we considered now the element $-U$:

$$V \rightarrow V' = (-U^\dagger)V(-U) = U^\dagger V U \quad (12)$$

So in general, we can $SU(2)$ the *double cover* of $SO(3)$, where each element transforms half as much as one might expect. This is the key concept behind "spinors", and the physics behind half integer spin particles.